# Mentoring Plan

### 1. Purpose and roles

The FI is the intellectual lead of the proposed research: the FI designed the project, authored this proposal, and will execute the analyses, write the papers as first author, and release the data products. The PI serves as research mentor and PI of record: scientific guidance, domain expertise, institutional accountability, and career development. This plan states how that relationship operates for the award's duration — a designed structure, not an assumption.

### 2. Mentoring structure and cadence

- Biweekly one-on-one meetings between FI and PI: progress against the timeline, technical roadblocks, and interpretation of results. The FI sets the agenda; the PI probes assumptions and connects results to the wider literature.
- Monthly research-group meetings: the FI presents work-in-progress to the group, building the habit of defending methods and results before an informed audience.
- Semester milestone reviews: at the start of each semester, FI and PI revisit the project timeline (§6 of the proposal) against actual progress, adjust scope where needed, and record the decisions. The honest-assessment section of the proposal (fallback plans per risk) is the standing agenda for these reviews.
- Committee alignment: the FI's dissertation committee reviews project progress at the institution's required milestones (qualifying exam, annual review, and proposal defense]), keeping the FINESST work and the degree requirements on track.

### 3. What the mentoring specifically develops

The proposal identifies two gaps in the FI's preparation; the mentoring plan is how these gaps are closed.

<u>Domain knowledge</u> — Antarctic geomorphology and hydrology. The FI's methods background is strong but was built on temperate-landscape problems. The PI will direct a structured reading program in cold-desert process geomorphology (energy-limited melt, permafrost/active-layer dynamics, closed-basin lake behavior),. By the end of Year 1, the FI can defend the physical interpretation of every Objective 1 driver to a polar audience, not just the statistics.

Statistical methodology and hierarchical modeling. The FI will complete graduate coursework in Bayesian/hierarchical statistical modeling before the Year-1 attribution analysis that depends on it. The PI will review model specifications at the design stage, before fitting — at that point the review will prevent mistakes instead of discovering them.

### 4. Professional development

- Conferences: the FI presents at AGU (cryosphere section) in Year 2 and at a SCAR/ISAES Antarctic venue in Year 3. Initially posters, then a hopefully an oral talk as results mature.

The PI introduces the FI to the McMurdo Dry Valleys research community — including the LTER network whose data this project uses — at these venues.

- Publication: three first-author papers are planned (proposal §6 timeline). The PI's role is manuscript review and journal strategy; the FI drafts, submits, and handles review responses, with the PI advising. Authorship follows institutional norms, stated in advance: the FI is first author on all project papers.
- Proposal craft: this proposal is itself a training artifact. During the award, the FI will draft at least one follow-on funding proposal with PI review — building the skills the fellowship strives to foster: becoming a fundable independent investigator.
- Open science practice: the FI maintains the public code repository, documentation, and data releases described in the OSDMP; the PI reviews each public release. This is treated as a professional skill under active mentorship, not an afterthought.
- Teaching/outreach: The FI will TA at least one semester of CIVE 3323 (Statistics for Engineers) and will also present his research in a long format (1 hour) research seminar class every fall semester.

**5. Career development**

- The FI and PI complete an Individual Development Plan (IDP) in the first semester of the award and revisit it annually: target career path, the skills and network that path needs, and identification of development goals for each project year.
- The PI commits to timely, candid letters and introductions as the FI approaches graduation, and to structuring Year 3 so that dissertation completion and the project's synthesis deliverables are aligned.

**6. Communication, feedback, and safeguards**

- Feedback: the PI provides written feedback on manuscripts and the annual NASA progress report within two weeks of receiving drafts. The FI flags potential issues early rather than at milestone reviews; the biweekly meeting exists so that no problem is more than a week old before the PI knows about it.
- Disagreement: scientific disagreements are resolved by evidence and recorded in the project's decision log. If the mentoring relationship itself needs mediation, the FI's dissertation committee and the department's chair are the named escalation path, per institutional policy.
- Availability: if the PI is on leave or travel for an extended period, Dr. Juan Carlos Fernandez-Diaz, NCALM PI, provides interim technical guidance so the FI is never unsupervised for more than two weeks.

**7. Assessment of the mentoring itself**

Once per year, FI and PI explicitly review this plan — what mentoring worked, what was missing, what the coming year's emphasis should be — and revise it. The plan is a working document, and will be treated as one.